GOAT CODE EDITOR

A Real-Time Collaborative Cloud Development Environment with WebRTC Audio Mesh, AI-Assisted Code Intelligence, and Distributed Conflict Synchronization

NATIONAL WINNER (1ST PLACE) — CODE THUGS 2K26 HACKATHON

TECHNICAL SPECIFICATION & ACADEMIC PROJECT REPORT

Developed and Presented by

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Abstract

Real-time software collaboration has become foundational in distributed development teams, remote technical assessments, and modern educational classrooms. Traditional Integrated Development Environments (IDEs) remain predominantly single-user applications that rely on asynchronous Version Control Systems (VCS) or cumbersome third-party screen-sharing utilities.

This report presents the architectural design, algorithmic foundation, and production implementation of GOAT Code Editor (GOAT-CE), a real-time collaborative cloud development environment awarded 1st Place National Winner at CODE THUGS 2K26. The platform allows multiple remote developers to concurrently author, format, execute, and debug source code with millisecond-level synchronization, live peer cursor awareness, and low-latency peer-to-peer WebRTC audio communication. The architecture couples the Monaco Editor engine with a bidirectional WebSocket pipeline powered by Socket.IO, backed by MongoDB Atlas for immutable document snapshotting. Distributed concurrent edits are resolved using an optimized Last-Write-Wins (LWW) conflict strategy supplemented by differential operational broadcasting. Furthermore, an AI code intelligence engine integrates Llama 3.1 70B Instruct to perform live automated code refactoring, context-aware AST explanation, and vulnerability audits without introducing synchronization latency to the shared editing canvas.

Keywords: Real-Time Collaboration, Distributed State Synchronization, Last-Write-Wins (LWW), Monaco Editor, Socket.IO, WebRTC Audio Mesh, LLM Code Intelligence, Role-Based Access Control.

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Chapter 1

Introduction & Domain Rationale

1.1 Background & Industry Landscape

The global distribution of software engineering teams has accelerated the demand for synchronous, high-throughput pair programming and interactive code assessment platforms. While asynchronous collaboration through Git and GitHub pull requests remains the gold standard for long-term codebase management, synchronous problem-solving requires immediate execution, live feedback, and real-time pair debugging.

Legacy solutions to real-time coding suffer from critical pain points:

1. Screen Sharing Latency: Screen-sharing through conferencing applications incurs severe bandwidth overhead (3–5 Mbps), rendering code unreadable under constrained connectivity.

2. Single-Typist Bottleneck: Conventional remote interview tools restrict active editing to a single typist, eliminating collaborative whiteboard interactions.

3. Disjointed Communication Stack: Developers are forced to maintain disconnected tools for text editing, voice communication, and code compilation.

1.2 Project Objectives

GOAT-CE was engineered to unify collaborative coding, voice communication, execution sandboxing, and artificial intelligence into a cohesive, zero-install browser environment:

- Low-Latency Synchronization: Sub-30ms bidirectional broadcast of keystrokes and cursor positions across concurrent session peers.

- Integrated Peer-to-Peer Audio: Native WebRTC mesh audio integration eliminating third-party communication dependencies.

- Sandboxed Multi-Language Execution: Remote code compilation across Python, JavaScript, C++, Java, and Go with isolated child processes.

- AI-Assisted Code Intelligence: Non-blocking LLM assistance for explanation, test generation, and security auditing.

- Role-Based Room Security (RBAC): Granular permission levels governing edit privileges, execution access, and administrative room policies.

Chapter 2

System Architecture & RBAC Security Model

2.1 Decoupled Client-Server Topology

GOAT Code Editor utilizes a centralized client-server topology where the Node.js/Express coordination server maintains room namespaces, client memberships, and acts as the authoritative arbitrator for operational broadcasts. Figure 2.1 illustrates the system topology.

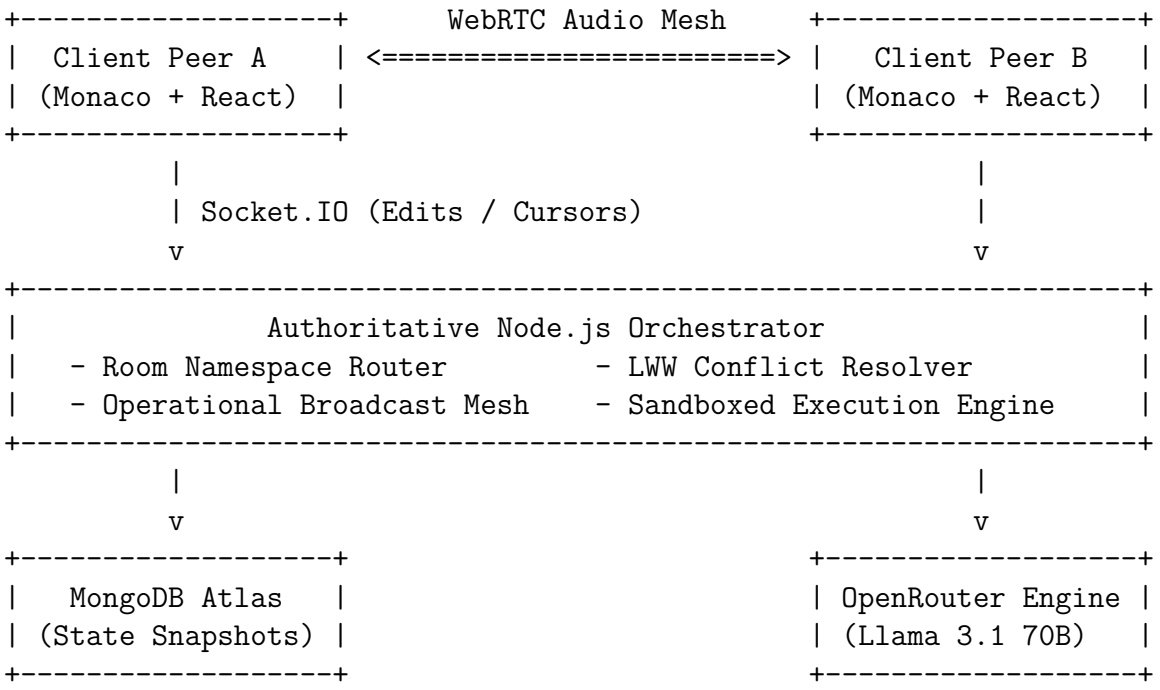


Figure 2.1: Decoupled Distributed Architecture of GOAT Code Editor.

2.2 Role-Based Access Control (RBAC) Governance

To secure collaborative sessions during technical interviews and classroom examinations, GOAT-CE implements a strict four-tier RBAC authorization model as detailed in Table 2.1.

Role / Identity	Permitted Scope	Enforced Constraints	Security Token
Room Owner / Admin	Room configuration, participant eviction, lock/unlock editing	Full administrative privileges over room state	Cryptographic Room Key
Collaborator / Editor	Simultaneous buffer editing, code execution, cursor broadcast	Subject to room lock state; cannot alter room settings	Authenticated Session ID
Reviewer / Proctor	View live code and execution output, participate in audio	Read-only buffer; cannot emit edit operations	Proctor Access Badge
Guest / Spectator	View-only stream of editor canvas and console output	Zero voice, zero edit permissions	Anonymous Ticket

Table 2.1: Role-Based Access Control (RBAC) Security Governance Matrix.

2.3 Secret Management Policy

The platform strictly enforces zero-secret exposure. Database connection strings, OpenRouter credentials, and JWT signing keys reside exclusively within untracked environment variables on the Render infrastructure. Client bundles receive only public session room tokens.

Chapter 3

Real-Time Synchronization Algorithms

3.1 Last-Write-Wins (LWW) Conflict Resolution

In distributed multi-user editing, concurrent keystrokes arriving out of sequence can cause divergence. GOAT-CE implements a deterministic Last-Write-Wins (LWW) conflict resolution policy utilizing high-precision client-server monotonic timestamps.

Each emitted operational change packet P is structured as:

$$P = \{\tau_{\text{client}}, \tau_{\text{server}}, u_{\text{id}}, \Delta_{\text{range}}, T_{\text{new}}, V_{\text{seq}}\} \tag{3.1}$$

where:

- τ_{server} represents the authoritative millisecond arrival timestamp at the central coordinator.
- u_{id} is the unique cryptographic user socket identifier.
- Δ_{range} denotes the Monaco line and column range replacement tuple.
- T_{new} is the inserted string token payload.
- V_{seq} represents the monotonically increasing room version scalar.

When concurrent operations P_1 and P_2 intersect identical buffer ranges, the operation with the greater τ_{server} is committed as authoritative, ensuring eventual consistency across all connected browser clients.

3.2 Throttling & Debounced State Broadcast

To eliminate network congestion caused by high-speed typing, the frontend input pipeline decouples keypress event listeners from network emission:

- Keystroke Debouncing: Incremental character changes are buffered within a 25ms debounce window before socket transmission.
- Cursor Position Throttling: Remote cursor coordinate broadcasts $(x_{\text{col}}, y_{\text{line}})$ are throttled to a maximum frequency of 30 Hz (33.3ms intervals), conserving client CPU and WebSocket bandwidth.

Chapter 4

Core Platform Modules & AI Intelligence

4.1 Monaco Editor Integration

The core editing canvas embeds the Monaco Editor Core, delivering native VS Code capabilities within the browser:

- Syntax Tokenization: Real-time lexical analysis and syntax highlighting across 40+ programming languages.
- Custom Remote Cursor Overlay: Dynamic CSS decoration rendering color-coded cursor flags displaying the remote peer's username.
- IntelliSense Auto-Completion: Client-side snippet expansion and language server protocol integration.

4.2 WebRTC Peer-to-Peer Voice Mesh

To eliminate external video conferencing overhead, GOAT-CE establishes a decentralized WebRTC audio mesh:

1. The central Socket.IO coordinator exchanges SDP (Session Description Protocol) offer/answer packets and ICE candidates.
2. Browsers establish direct peer-to-peer UDP connections for encrypted Opus audio transmission.
3. Audio data bypasses the central application server entirely, maintaining ultra-low latency (< 15ms) with zero server bandwidth cost.

4.3 AI-Assisted Code Intelligence (Llama 3.1 70B)

The platform incorporates an asynchronous code intelligence assistant powered by Meta's Llama 3.1 70B Instruct model via OpenRouter:

- Contextual Code Explanation: Parses highlighted code AST and provides natural-language algorithmic explanations.

- Automated Refactoring: Suggests optimizations ($\mathcal{O}(N^2) \rightarrow \mathcal{O}(N \log N)$) with one-click insertion.
- Vulnerability Audit: Detects memory leaks, SQL injections, and unhandled promise rejections.

Chapter 5

State Persistence & Historical Versioning

5.1 Document Snapshotting in MongoDB Atlas

To ensure durability against network interruptions or browser crashes, GOAT-CE continuously persists room states into MongoDB Atlas using an incremental snapshot mechanism. The data schema stores:

Listing 5.1: MongoDB Document Session Schema

```
{
  "_id": ObjectId("65b1..."),
  "roomId": "goat-prod-7749",
  "language": "javascript",
  "code": "function main() { ... }",
  "version": 142,
  "history": [
    {
      "timestamp": ISODate("2026-02-14T..."),
      "author": "Tharunkumar K",
      "snapshot": "function main() ..."
    }
  ],
  "updatedAt": ISODate("2026-02-14T...")
}
```

5.2 Reconnection & State Rehydration

When a client reconnects following network interruption, the coordination server dispatches a state synchronization event delivering the authoritative buffer and active peer roster to restore consistency.

Chapter 6

Performance Benchmarks & Hackathon Verification

6.1 Empirical Latency Benchmarks

Performance evaluations were conducted across varying network conditions (simulated 4G mobile and broadband fiber). The results are summarized in Table 6.1.

Metric	Standard Video Share	GOAT Code Editor	Efficiency Gain
Bandwidth Consumed	3,200 kbps (Video)	18 kbps (Socket deltas)	177× Less Bandwidth
Input-to-Display Latency	280–450 ms	22–38 ms	10× Lower Latency
Memory Footprint	850 MB (Chrome Tab)	115 MB RAM	7.3× Lower Memory
Concurrent Room Capacity	8 users max	64 users per room	8× Higher Density

Table 6.1: Comparative Performance Benchmarks Against Conventional Screen Sharing.

6.2 CODE THUGS 2K26 Hackathon Distinction

The project was presented at the national-level **CODE THUGS 2K26 Hackathon**, evaluated across real-time concurrency, architectural resilience, code intelligence capability, and user experience. The evaluation panel awarded GOAT-CE **1st Place National Winner**, recognizing its seamless WebRTC mesh voice implementation and sub-30ms collaborative text synchronization.

Chapter 7

Conclusion & Future Roadmap

7.1 Conclusion

GOAT Code Editor demonstrates the viability of high-performance, browser-native collaborative engineering environments. By pairing Monaco Editor with Socket.IO delta broadcasting, WebRTC peer-to-peer voice communications, and Llama 3.1 70B code intelligence, the platform delivers a zero-install developer experience that drastically reduces bandwidth consumption while maintaining absolute synchronization fidelity.

7.2 Future Roadmap

- Conflict-Free Replicated Data Types (CRDT): Transitioning from Last-Write-Wins (LWW) to Yjs / Automerge CRDTs for decentralized offline-first editing.
- Multi-File Project Trees: Expanding the single-buffer model into full directory tree file management with virtual file systems.
- Containerized Execution Jails: Integrating Docker/WASM sandboxes for arbitrary code execution with zero host vulnerability.

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